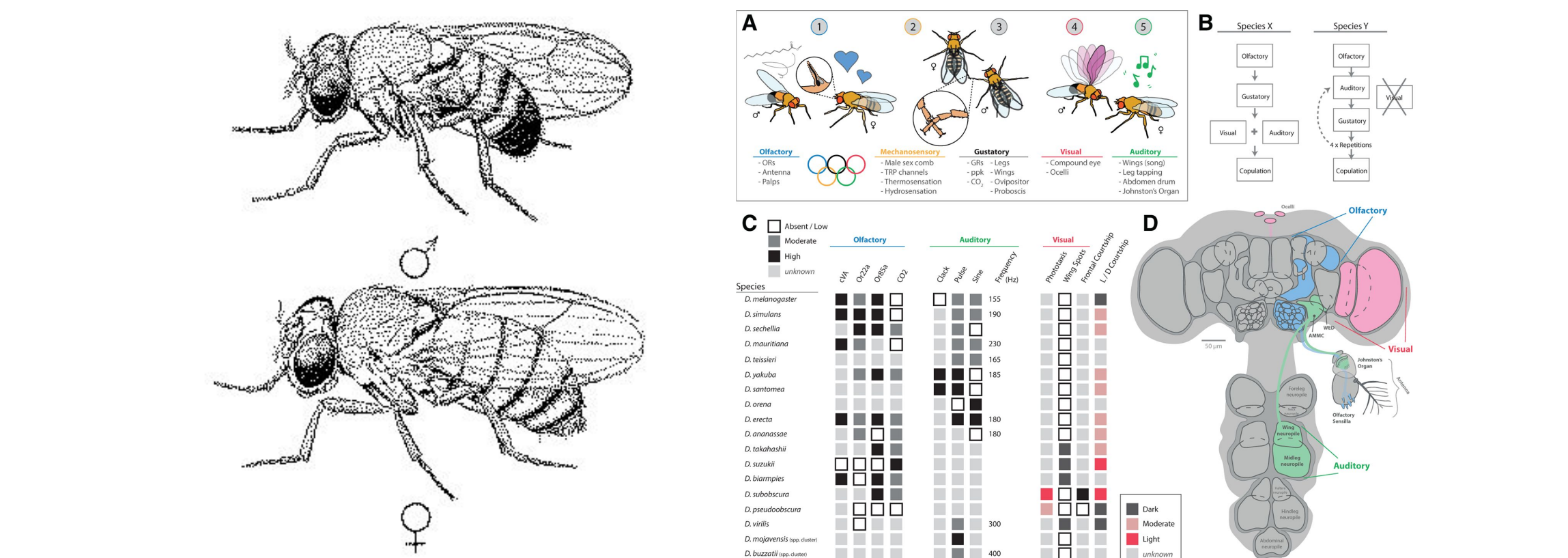


Background

Ecological specialization plays an important role in shaping how different species interact with their environment through their sensory systems that detect chemical cues. In Drosophilidae, gustatory receptor (GR) genes are responsible for detecting sugars, toxins, and other compounds that influence feeding behavior and habitat selection. ***This study investigates whether ecological specialization predicts differences in GR gene repertoire size, while understanding how environmental chemical pressures shape sensory evolution.***



In the Drosophilidae family, different species occupy different ecological niches, from generalists that feed on many types of substrates to specialists that rely on specific resources. These ecological differences expose species to varying chemical environments, which can impose selective pressures on sensory genes. A key question in evolutionary biology is whether these differences are driven by ecological adaptation or by shared evolutionary history among closely related species.

Study Design & AI

The GR gene counts and ecological classifications were analyzed across multiple species via a phylogenetic comparative framework. A Phylogenetic Generalized Least Squares (PGLS) model was applied to account for shared evolutionary history. The results indicate that ecological specialization is associated with variation in GR gene repertoire size, suggesting that species adapted to different ecological niches experience distinct selective pressures on their sensory systems. These findings suggest that environmental chemical cues drive the evolution of sensory gene families.

A total of 17 species were included in the analysis. Data were standardized to remove duplicates and inconsistencies. Artificial intelligence tools were used to assist with data organization and interpretation, while all biological reasoning and final analytical decisions were validated independently.

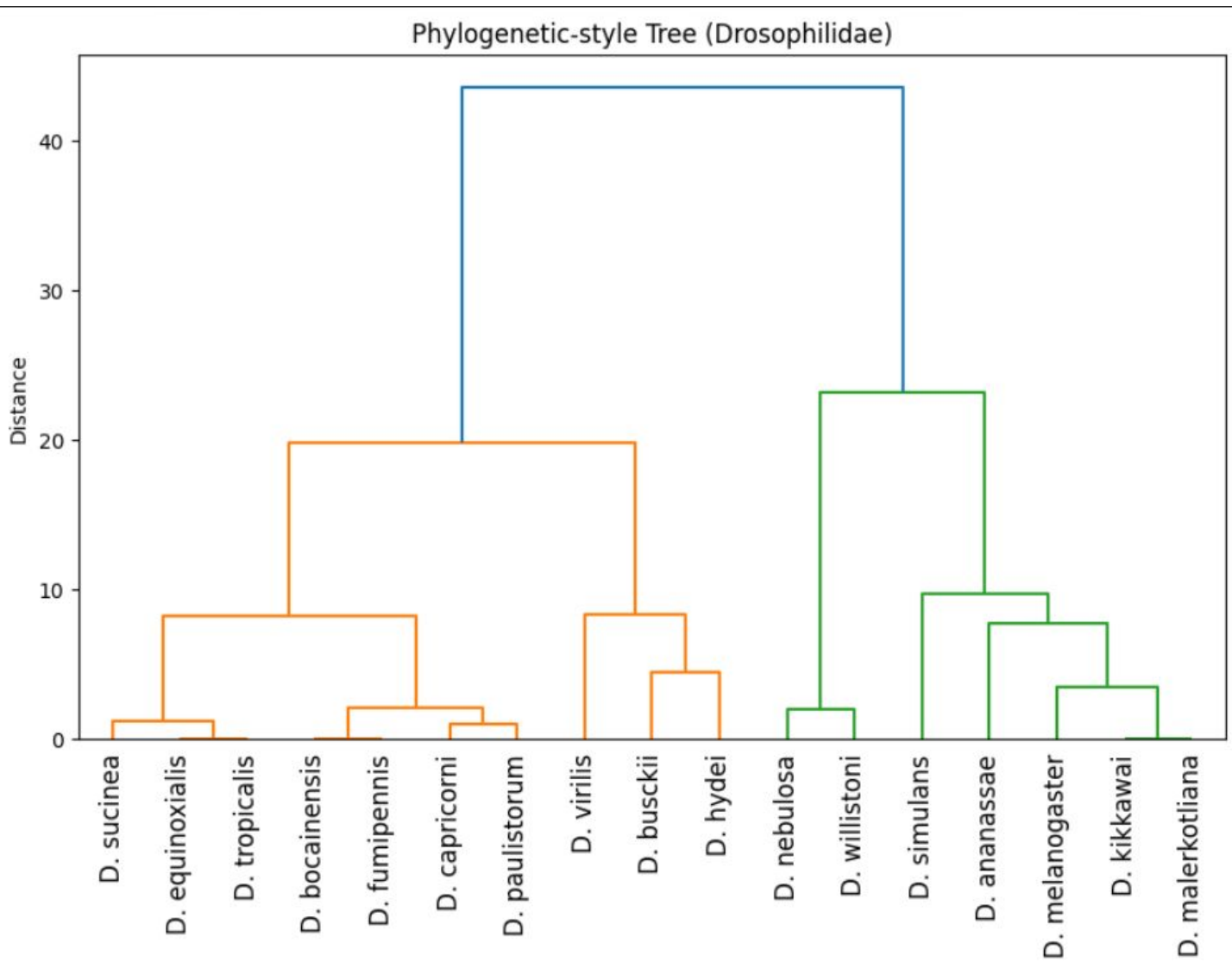
To ensure accurate evolutionary inference, all species data were analyzed in the context of their phylogenetic relationships.

Model Fitting

To test the relationship between ecological specialization and GR gene repertoire size, a Phylogenetic Generalized Least Squares (PGLS) model was used. This approach incorporates the phylogenetic tree into the regression framework, allowing the analysis to account for shared ancestry among species.

The model examined GR gene count as a function of host breadth while accounting for phylogenetic relationships among species. By including phylogeny, the analysis reduces the influence of shared ancestry which in hand allows for a more accurate assessment of the relationship between ecological specialization and GR gene repertoire size. This approach helps distinguish whether observed patterns are due to evolutionary history or ecological factors.

Figure 1. Phylogenetic-style clustering of Drosophilidae species based on host breadth and GR gene repertoire size. GR gene counts are annotated for each species. Species that cluster together tend to share similar GR counts, indicating that variation in GR repertoire size is strongly influenced by shared evolutionary history rather than independent ecological adaptation.



- A positive relationship was initially observed between host breadth and GR gene repertoire size, with more generalist species tending to have higher GR counts. This trend is visible in the scatterplot, where GR gene count increases with the number of plant orders utilized.
- However, after applying the PGLS model and accounting for phylogenetic relationships, this relationship was no longer statistically significant. This indicates that the apparent trend is largely driven by shared ancestry rather than independent ecological adaptation. In other words, closely related species tend to have similar GR gene counts regardless of differences in host breadth.
- These results suggest that phylogeny plays a dominant role in shaping GR gene repertoire size in Drosophilidae, limiting the direct impact of ecological specialization on this trait.
- GR counts come from genome sequencing studies where researchers identify and count gustatory receptor genes using annotation pipelines.

Results

Species Data Summary

Species	Host Breadth (Plant Orders)	GR Gene Count
<i>D. busckii</i>	8	34
<i>D. ananassae</i>	10	54
<i>D. kikkawai</i>	11	60
<i>D. mallerkotlia na</i>	11	60
<i>D. melanogaster</i>	14	60
<i>D. simulans</i>	19	60
<i>D. hydei</i>	6	38
<i>D. virilis</i>	1	40
<i>D. bocainensis</i>	6	47
<i>D. capricorni</i>	8	47
<i>D. equinoxialis</i>	2	47
<i>D. fumipennis</i>	6	47
<i>D. nebulosa</i>	19	47
<i>D. paulistorum</i>	7	47
<i>D. sucinea</i>	3	47
<i>D. tropicalis</i>	2	47
<i>D. willistoni</i>	21	47

Figure 2. Summary of Drosophilidae species included in this study, showing host breadth (measured as the number of plant orders utilized) and corresponding GR gene repertoire size. These data were used to examine the relationship between ecological specialization and sensory gene evolution.

we use plant host breadth because fruit flies depend on plant-based substrates for feeding and reproduction, so it reflects the diversity of chemical environments they experience.

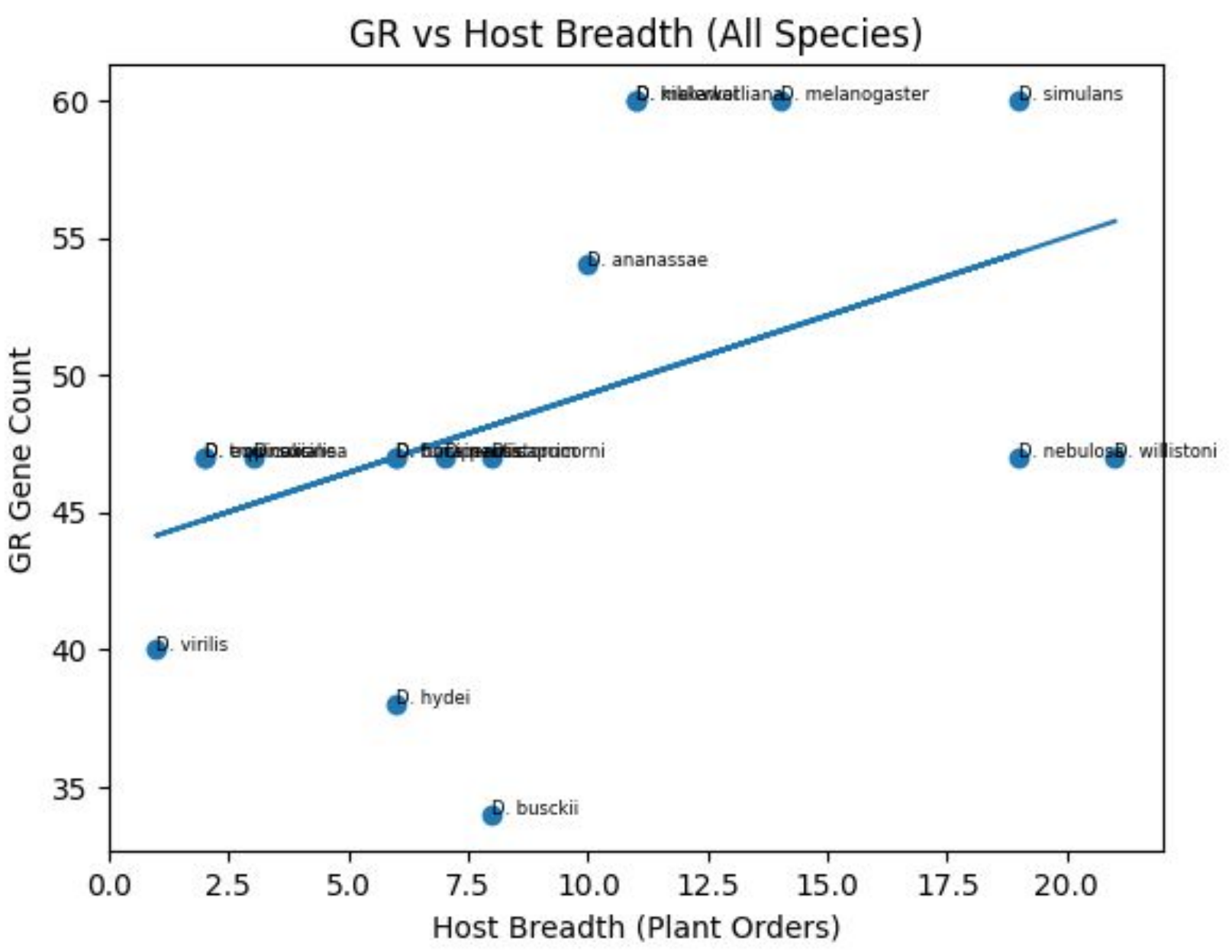


Figure 3. A positive relationship was observed between host breadth and GR gene repertoire size in Drosophilidae, with more generalist species tending to have higher GR counts. However, after accounting for phylogeny, this relationship was no longer statistically significant, indicating that the pattern is largely driven by shared ancestry rather than independent ecological adaptation.

Results Analysis and Discussion

Discussion

- The pattern that was observed revealed the importance of accounting for phylogeny in comparative studies. While the raw data suggested that generalist species have larger GR gene repertoires, this relationship weakens when evolutionary traits are considered. This implies that GR gene counts are not evolving independently in response to ecological specialization, but instead reflect within the inherited traits.
- One possible explanation is that GR gene repertoire size is relatively conserved within evolutionary clades with limited ability for adaptation in environmental changes. Alternatively, adaptation to specific host environments may occur through changes in gene expression or receptor sensitivity rather than changes in gene number.
- These findings emphasize that ecological patterns observed in raw data can be misleading if phylogenetic relationships are not considered.

Key Results

- A positive relationship was observed between host breadth and GR gene repertoire size in the raw data.
- More generalist species tended to have higher GR gene counts.
- After applying PGLS, the relationship was no longer statistically significant.
- The observed pattern is primarily driven by shared ancestry rather than independent ecological adaptation.
- GR gene repertoire size appears to be conserved within evolutionary lineages.

Overall, these results show that ecological trends in GR gene repertoire size are mainly shaped by evolutionary history which highlight the importance of accounting for phylogeny when interpreting patterns of sensory evolution.

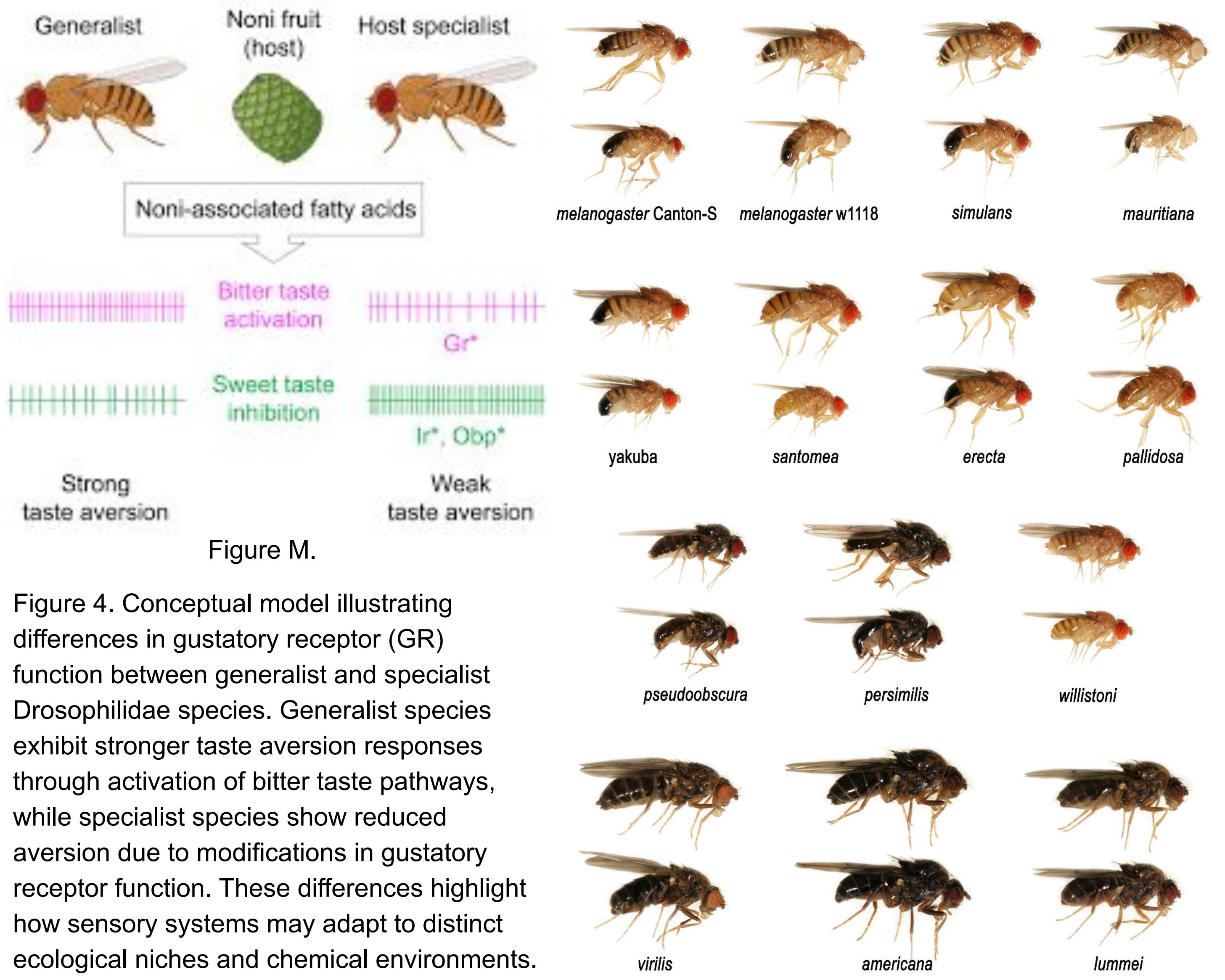


Figure M. Figure 4. Conceptual model illustrating differences in gustatory receptor (GR) function between generalist and specialist Drosophilidae species. Generalist species exhibit stronger taste aversion responses through activation of bitter taste pathways, while specialist species show reduced aversion due to modifications in gustatory receptor function. These differences highlight how sensory systems may adapt to distinct ecological niches and chemical environments.

Conclusions & Acknowledgements

This study demonstrates that while there is an apparent positive relationship between host breadth and GR gene repertoire size in Drosophilidae, this pattern is not statistically significant after accounting for phylogeny. These results suggest that shared ancestry, rather than ecological specialization, is the primary driver of variation in GR gene counts across species. Overall, this work highlights the importance of phylogenetic comparative methods in distinguishing true ecological effects from evolutionary background patterns and provides insight into how sensory gene evolution is shaped across lineages. In my future work I will attempt to expand sampling as my small sample size may also be contributing to my ability to detect an impact of ecological specialization.